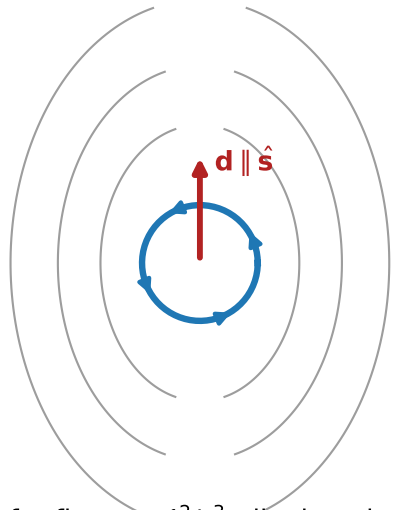
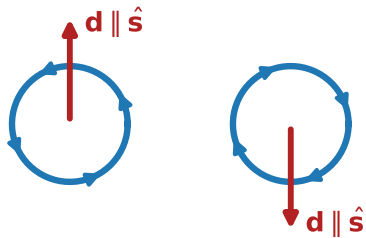


(a) one electron $\sim \ell$ (fm)



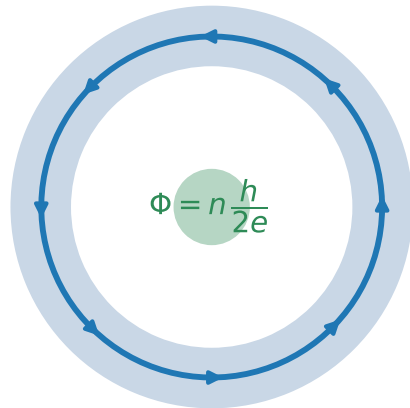
far flow $\sim \kappa \ell^2 / r^3$: dipole only

(b) Cooper pair $\sim \xi$ (40 nm)



$\mathbf{d}_1 + \mathbf{d}_2 \propto \hat{\mathbf{S}}_{\text{tot}} \Rightarrow 0$ on the singlet

(c) ring $\sim$ mm: 10^{23} dark pairs, one phase



electronic phase winds: $\oint \nabla \theta \cdot d\mathbf{l} = 2\pi n$